



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO SSM AUTO SERVICE

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

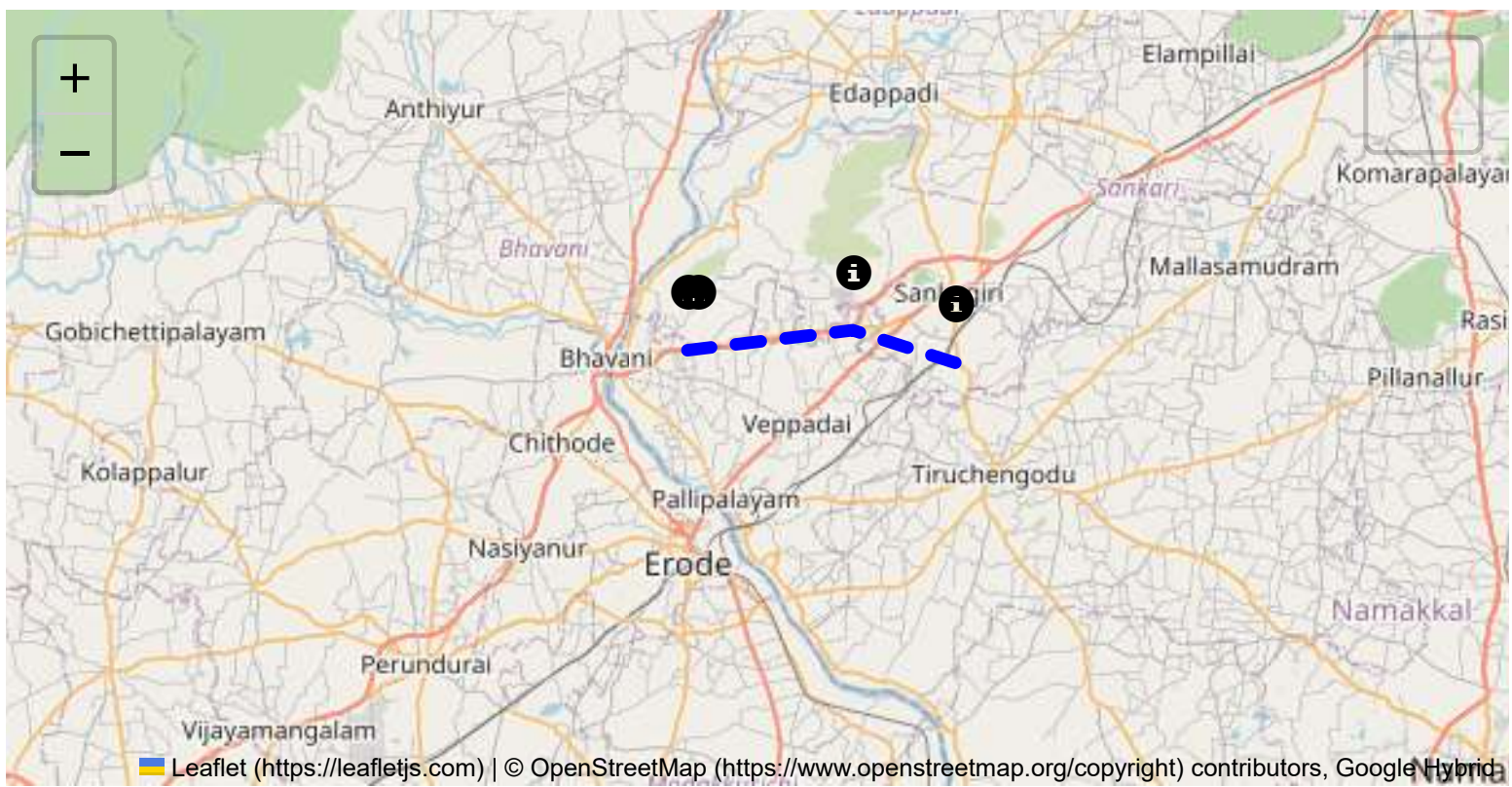
## **Journey Risk Mitigation**

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

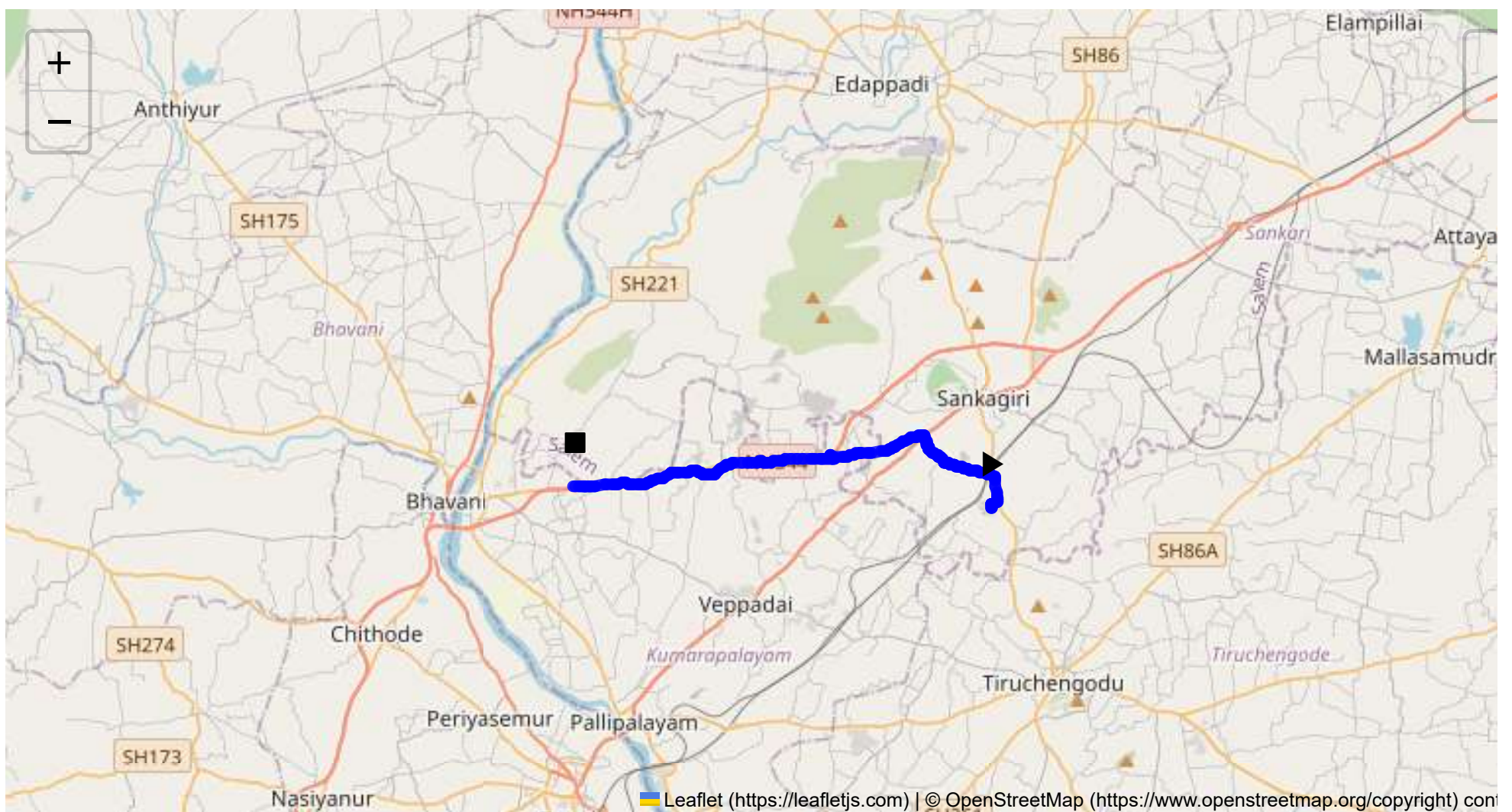
## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 18.86 km**  
**Estimated Duration: 0.4 hours**  
**Adjusted Duration (Heavy Vehicle): 0.5 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.445575, 77.726959)**



## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

The route between CVQF+23W, Sangagiri to CPWG+6XM, Jothi Nagar, covers approximately 18.86 kilometers, passing through Puthur, Padaiveedu, and Tiruchengode. This route involves traveling on local roads that connect to significant highways, including segments on the Salem - Kochi Highway.

## 2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu tends to have a tropical climate with hot summers and moderate monsoons. The major weather-related hazards along the route include:

- **Monsoon (June to September):** Heavy rains can lead to waterlogging and reduced visibility.
- **Summer (March to June):** High temperatures might cause road fatigue or tire issues.

## 3. Analysis of Traffic Patterns

- **Peak Hours:** Traffic tends to be heavier in the mornings (8-10 AM) and evenings (5-8 PM) due to local commuting.
- **Congestion-Prone Areas:** Town centers in Tiruchengode and Sangagiri may experience congestion, especially near marketplaces and local junctions.

## 4. Assessment of Road Quality and Infrastructure

- **Sangagiri to Puthur:** Generally well-maintained roads, but narrow in some areas.
- **Puthur to Padaiveedu:** Infrastructure may include some potholes or uneven surfaces, common in rural road sections.
- **Padaiveedu to Tiruchengode:** Better-maintained roads; however, sharp turns and local traffic might be problematic.
- **Overall:** Mixed road quality; regular upkeep is necessary, especially after monsoons.

## 5. Suggestions for Alternative Routes for Emergencies

- **Route Diversion:** In case of obstruction, consider local roads that reconnect to primary state highways.
- **Detours:** Use digital maps for real-time alternate route suggestions.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

- **Permits:** Ensure all necessary permits are obtained for transporting hazardous materials.
- **Timing Regulations:** Follow any restrictions on heavy vehicle transit during peak hours.

## 7. Overview of Historical Incidents

There have been few high-profile incidents involving heavy vehicles in this region, but common issues include:

- **Overloading incidents:** Leading to accidents due to mechanical failure.
- **Monsoon accidents:** Slippage and hydroplaning are risks during monsoon seasons.



## 8. Environmental Considerations and Sensitive Areas

- Protected Areas:** Minimal impact on direct routes, but nearby fields and water bodies may present environmental spill risks.
- Waste Disposal:** Prohibited in certain zones; adhere to local guidelines rigorously.

## 9. Communication Coverage

- Coverage Check:** Generally good across the route with few potential dead zones outside of urban areas.
- Network Providers:** Ensure devices have access to multiple network providers to reduce the risk of communication failures.

## 10. Estimated Emergency Response Times

- Sangagiri to Puthur:** 15-25 minutes due to proximity to local towns.
- Puthur to Tiruchengode:** 25-35 minutes, varying with road conditions.
- Tiruchengode to Jothi Nagar:** 20-30 minutes, longer in congested conditions.

## 12. Overall Summary of Risk Assessment

This route has moderate risk factors primarily due to:

- Weather Conditions:** Rain and heat pose notable and variable risks.
- Traffic Patterns:** Regular updates on traffic congestion are advisable.
- Road Infrastructure:** Maintenance varies significantly; vigilance is necessary to avoid road hazards.
- Preparation:** Adequate emergency preparedness and route planning are crucial to mitigate potential risks effectively.

Traveling on this route with hazardous materials necessitates caution, thorough planning, and vigilance, especially during inclement weather or peak traffic periods.

### Risk Assessment - Turns

|   | Risk Type | Risk Level | Coordinates        | Speed Limit |
|---|-----------|------------|--------------------|-------------|
| 0 | Turn      | High       | 11.43811, 77.87348 | 15 KM/Hr    |
| 1 | Turn      | Medium     | 11.43961, 77.87341 | 30 KM/Hr    |
| 2 | Turn      | Medium     | 11.43971, 77.87348 | 30 KM/Hr    |
| 3 | Turn      | High       | 11.44029, 77.87544 | 15 KM/Hr    |
| 4 | Turn      | Medium     | 11.44898, 77.87410 | 30 KM/Hr    |

|    | Risk Type  | Risk Level | Coordinates        | Speed Limit |
|----|------------|------------|--------------------|-------------|
| 5  | Turn       | Medium     | 11.45348, 77.85706 | 30 KM/Hr    |
| 6  | Turn       | Medium     | 11.46318, 77.84913 | 30 KM/Hr    |
| 7  | Turn       | Medium     | 11.45697, 77.82719 | 30 KM/Hr    |
| 8  | Turn       | High       | 11.45542, 77.81725 | 15 KM/Hr    |
| 9  | Blind Spot | Blind Spot | 11.45631, 77.81668 | 10 KM/Hr    |
| 10 | Turn       | High       | 11.45636, 77.81678 | 15 KM/Hr    |
| 11 | Blind Spot | Blind Spot | 11.45543, 77.81740 | 10 KM/Hr    |
| 12 | Turn       | High       | 11.45516, 77.81346 | 15 KM/Hr    |
| 13 | Turn       | High       | 11.45521, 77.81343 | 15 KM/Hr    |
| 14 | Blind Spot | Blind Spot | 11.44546, 77.73070 | 10 KM/Hr    |
| 15 | Blind Spot | Blind Spot | 11.44574, 77.73071 | 10 KM/Hr    |

## Emergency Locations

|   | type     | name                                 | coordinates            | speed_limit | risk_level |
|---|----------|--------------------------------------|------------------------|-------------|------------|
| 3 | hospital | Government Hospital                  | 11.4525596, 77.7749426 | 30 km/h     | Medium     |
| 4 | hospital | Dhanvantri Multi Speciality Hospital | 11.451362, 77.766602   | 30 km/h     | Medium     |
| 5 | hospital | Dhanvanthri Hospital                 | 11.4496712, 77.7593772 | 30 km/h     | Medium     |
| 6 | hospital | J.K.K. Trust Hospital                | 11.4445841, 77.7307962 | 30 km/h     | Medium     |

## Crowded Spots

|   | type        | name                             | coordinates            | speed_limit | risk_level |
|---|-------------|----------------------------------|------------------------|-------------|------------|
| 0 | school      | KRP Matric. Hr. Sec School       | 11.4546193, 77.8142445 | 30 km/h     | Medium     |
| 1 | college     | Vivekanandha Engineering College | 11.4589312, 77.7899284 | 30 km/h     | Medium     |
| 2 | marketplace | Monday market                    | 11.452863, 77.775989   | 30 km/h     | Medium     |

## Route Photos of Risky Spots





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44029, 77.87544



**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.44898, 77.87410





**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.45348, 77.85706**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.46318, 77.84913**



**Risk Type: Turn**

**Risk Level: Medium**

**Speed Limit: 30 KM/Hr**

**Coordinates: 11.45697, 77.82719**





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.45542, 77.81725



**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.45631, 77.81668





**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.45636, 77.81678**



**Risk Type: Blind Spot**

**Risk Level: Blind Spot**

**Speed Limit: 10 KM/Hr**

**Coordinates: 11.45543, 77.81740**



**Risk Type: Turn**

**Risk Level: High**

**Speed Limit: 15 KM/Hr**

**Coordinates: 11.45516, 77.81346**





**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.45521, 77.81343



**Risk Type:** Blind Spot

**Risk Level:** Blind Spot

**Speed Limit:** 10 KM/Hr

**Coordinates:** 11.44546, 77.73070



**Risk Type:** Blind Spot  
**Risk Level:** Blind Spot  
**Speed Limit:** 10 KM/Hr  
**Coordinates:** 11.44574, 77.73071

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